SHEAFIFYING THE DIFFERENTIALS OF A MORPHISM OF SCHEMES ON THE SMALL ETALE SITE GIVES THE MODULE OF DIFFERENTIALS OF THE MORPHISM OF ETALE SITES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-more-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the first line of the pinned chapter file is \input{preamble}, and the native preamble was not staged with the source, so no preamble text lives inside the chapter. The retained preamble used here is the project's single Stacks-wide preamble (the same 217-line / 8116-byte file that the already published chow and etale-cohomology cards use), copied mechanically line for line from sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointers are located at their \label lines inside the pinned chapter file; the cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The proof is complete except for the last sentence: the author writes 'We omit the verification' of the two identities showing c" and c mutually inverse on local sections (proof line 941). The construction of both maps and the adjunction argument are inside the counted span, so the omission is a routine tail, not a missing argument.

Allowed standard prerequisites (editorial).

- The sheaf of differentials is insensitive to etale localization. More on Morphisms, Lemma labelled more-morphisms-lemma-sheaf-differentials-etale-localization; invoked at lines 881-883. Gives h^*Omega_{X/Y} = Omega_{U/Y} for an etale h : U -> X, which identifies the values of (Omega_{X/Y})^a.
- Universal property of the module of differentials on a ringed site. Modules on Sites, Definition labelled sites-modules-definition-module-differentials; invoked at lines 891-898 and 912-913. Produces the universal O_{X_etale}-linear map c out of the derivation d^a.
- Comparison of the small Zariski and small etale sites of a scheme. Descent, Remark labelled descent-remark-change-topologies-ringed and Lemma labelled descent-lemma-compare-sites; invoked at lines 920-927. Realises the restriction to the small Zariski site as the pushforward along id_{small,etale,Zar}, so the map c' has an adjoint c".
- Sheaves on the small Zariski site of a scheme are sheaves on the scheme. Descent, Remark labelled descent-remark-Zariski-site-space; invoked at lines

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

905-907. Lets a derivation on the etale site be restricted to an ordinary Y-derivation on X.

Source reference. Original lemma 070S: sources/stacks-chow/tags.txt, source lines 8966–8966. Navigation pointer for \ref{descent-lemma-compare-sites} used inside the counted span (line 924); the argument that consumes it is retained in the span.

Source reference. Original lemma 03FG: sources/stacks-chow/tags.txt, source lines 4491–4491. Navigation pointer for \ref{descent-remark-Zariski-site-space} used inside the counted span (line 907); the argument that consumes it is retained in the span.

Source reference. Original lemma 070R: sources/stacks-chow/tags.txt, source lines 8965–8965. Navigation pointer for \ref{descent-remark-change-topologies-ringed} used inside the counted span (line 865); the argument that consumes it is retained in the span.

Source reference. Original lemma 04BX: sources/stacks-chow/tags.txt, source lines 5669–5669. Navigation pointer for \ref{more-morphisms-lemma-sheaf-differentials-etale-localization} used inside the counted span (line 883); the argument that consumes it is retained in the span.

Source reference. Original lemma 01UQ: sources/stacks-chow/tags.txt, source lines 2419–2419. Navigation pointer for \ref{morphisms-definition-sheaf-differentials} used inside the counted span (line 873); the argument that consumes it is retained in the span.

Source reference. Original lemma 04BN: sources/stacks-chow/tags.txt, source lines 5658–5658. Navigation pointer for \ref{sites-modules-definition-module-differentials} used inside the counted span (line 876); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $f_{small} : X_{étale} \to Y_{étale}$ be the associated morphism of small étale sites, see Descent, Remark 070R. Then there is a canonical isomorphism*

$$(\Omega_{X/Y})^a = \Omega_{X_{étale}/Y_{étale}}$$

compatible with universal derivations. Here the first module is the sheaf on $X_{étale}$ associated to the quasi-coherent $\mathcal{O}_X$-module $\Omega_{X/Y}$, see Morphisms, Definition 01UQ, and the second module is the one from Modules on Sites, Definition 04BN.

Proof. Let $h : U \to X$ be an étale morphism. In this case the natural map $h^*\Omega_{X/Y} \to \Omega_{U/Y}$ is an isomorphism, see More on Morphisms, Lemma 04BX. This means that there is a natural $\mathcal{O}_{Y_{étale}}$-derivation

$$\mathrm{d}^a : \mathcal{O}_{X_{étale}} \longrightarrow (\Omega_{X/Y})^a$$

since we have just seen that the value of $(\Omega_{X/Y})^a$ on any object U of $X_{étale}$ is canonically identified with $\Gamma(U, \Omega_{U/Y})$. By the universal property of $\mathrm{d}_{X/Y} : \mathcal{O}_{X_{étale}} \to \Omega_{X_{étale}/Y_{étale}}$ there is a unique $\mathcal{O}_{X_{étale}}$-linear map $c : \Omega_{X_{étale}/Y_{étale}} \to (\Omega_{X/Y})^a$ such that $\mathrm{d}^a = c \circ \mathrm{d}_{X/Y}$.

Conversely, suppose that $\mathcal{F}$ is an $\mathcal{O}_{X_{étale}}$-module and $D : \mathcal{O}_{X_{étale}} \to \mathcal{F}$ is a $\mathcal{O}_{Y_{étale}}$-derivation. Then we can simply restrict D to the small Zariski site X_{Zar} of X. Since sheaves on X_{Zar} agree with sheaves on X, see Descent, Remark 03FG, we see that $D|_{X_{Zar}} : \mathcal{O}_X \to \mathcal{F}|_{X_{Zar}}$ is just a "usual" Y-derivation. Hence we obtain a map $\psi : \Omega_{X/Y} \longrightarrow \mathcal{F}|_{X_{Zar}}$ such that $D|_{X_{Zar}} = \psi \circ \mathrm{d}$. In particular, if we apply this with $\mathcal{F} = \Omega_{X_{étale}/Y_{étale}}$ we obtain a map

$$c' : \Omega_{X/Y} \longrightarrow \Omega_{X_{étale}/Y_{étale}}|_{X_{Zar}}$$

Consider the morphism of ringed sites $\mathrm{id}_{small,étale,Zar} : X_{étale} \to X_{Zar}$ discussed in Descent, Remark 070R and Lemma 070S. Since the restriction functor $\mathcal{F} \mapsto \mathcal{F}|_{X_{Zar}}$ is equal to $\mathrm{id}_{small,étale,Zar,*}$, since $\mathrm{id}^*_{small,étale,Zar}$ is left adjoint to $\mathrm{id}_{small,étale,Zar,*}$ and since $(\Omega_{X/Y})^a = \mathrm{id}^*_{small,étale,Zar}\Omega_{X/Y}$ we see that c' is adjoint to a map

$$c'' : (\Omega_{X/Y})^a \longrightarrow \Omega_{X_{étale}/Y_{étale}}.$$

We claim that c'' and c' are mutually inverse. This claim finishes the proof of the lemma. To see this it is enough to show that $c''(\mathrm{d}(f)) = \mathrm{d}_{X/Y}(f)$ and $c(\mathrm{d}_{X/Y}(f)) = \mathrm{d}(f)$ if f is a local section of $\mathcal{O}_X$ over an open of X. We omit the verification. $\square$